

VINAY PARTAP SINGH

bhardwajvinay400@gmail.com | +91-8295262875 [LinkedIn](#) | [GitHub](#) | LeetCode: 300+ | CN: 225+

PROFESSIONAL SUMMARY

Full Stack Software Engineer with production-grade experience building microservice architectures, LLM-powered platforms, REST APIs, and AI-integrated web applications. Proficient in React.js, Node.js, FastAPI, LangGraph, PostgreSQL, Redis, and AWS. Designed and shipped DevBattles — an AI coding assessment platform with 25+ APIs, LangGraph evaluation pipeline, JWT/RBAC security, and automated CI/CD on AWS EC2. Strong foundation in Data Structures & Algorithms (LeetCode 290+) and system design. Seeking an SDE role to build scalable, secure, and impactful software products.

TECHNICAL SKILLS

Languages	Java, Python, JavaScript (ES6+), TypeScript, C++, SQL
Frontend	React.js, Next.js, HTML5, CSS3, Responsive Design, REST API Integration
Backend	Node.js, Express.js, FastAPI, REST APIs, JWT, RBAC, Rate Limiting, MVC Architecture
AI / LLM	LangGraph, Google Gemini API, Prompt Engineering, YOLOv8, PyTorch, OpenCV
Databases	PostgreSQL, MongoDB, Redis, MySQL, Neon, ACID Transactions, Schema Design
DevOps / Cloud	AWS (EC2, S3, IAM), Docker, Nginx, PM2, GitHub Actions, CI/CD, Vercel, Render
Concepts	Microservice Architecture, System Design, OOP, SDLC, Agile, Secure Auth

PROJECTS

DevBattles — AI Coding Assessment Platform [\[Live\]](#) [\[GitHub\]](#) 2025 – 2026

React.js, Node.js, Express, FastAPI, LangGraph, PostgreSQL, Redis, AWS EC2, Nginx, PM2, GitHub Actions

- Architected and deployed an AI-powered coding assessment platform using a 3-service microservice architecture (React frontend, Node.js/Express backend, FastAPI AI service) with automated CI/CD using GitHub Actions, PM2, and Nginx on AWS EC2
- Developed 25+ REST APIs for authentication, coding contests, homework, submissions, user management, and AI evaluation — implemented JWT authentication, role-based access control (RBAC), and IP/user-based rate limiting for secure and scalable API access
- Built an LLM-driven code evaluation pipeline using LangGraph to analyze programming submissions, compare against expected solutions, generate personalized feedback, and support automated assessment across multiple programming languages

VisionGuard AI — Computer Vision System [\[Live\]](#) [\[GitHub\]](#) 2025

Python, YOLOv8, PyTorch, OpenCV, Streamlit, Streamlit Cloud

- Built real-time pedestrian and vehicle detection system using YOLOv8 achieving ~95% accuracy at ~15 FPS with proximity-based risk classification (HIGH/MEDIUM/LOW) and cooldown-controlled audio alerts reducing false alert rate by ~70%
- Engineered multi-feature CV pipeline: centroid-based object tracking with persistent IDs, pixel-displacement speed estimation (km/h), custom polygon danger zones with ray-casting hit detection, and Gaussian heatmap accumulator with temporal decay
- Deployed full-stack application on Streamlit Cloud with role-based login (Admin/Viewer), 15-day road safety report engine scoring roads 0–100 across 3 risk tiers (SAFE/CAUTION/AVOID), and session-persistent analytics with CSV/report export

EXPERIENCE

Software Developer Intern - Infosys Springboard

July 2026 | Remote

- Developed a **full-stack Government Policy & Scheme Management Portal** using the **MERN Stack**, building **20+ responsive pages** and **3 role-based dashboards** (Citizen, Government Official & Admin).
- Engineered **15+ reusable React components**, implemented **JWT authentication**, protected routing, and role-based access control, improving UI consistency across the application.
- Collaborated in an Agile team using **Git/GitHub**, contributing to **10+ core modules** including Policy Management, Scheme Management, Approval Workflow, Reports, and Notifications.

Pantech.AI — AI/ML Intern

Feb 2025 | Remote

- Completed end-to-end AI/ML project development covering model building, data preprocessing, training pipelines, and Python-based production deployment

EDUCATION

K.R. Mangalam University, Gurgaon

2023 – 2027 (Expected)

B.Tech — Computer Science Engineering | **CGPA: 8.9 / 10** | **Dean's List Scholar**

Coursework: Data Structures & Algorithms, Operating Systems, DBMS, Computer Networks, Software Engineering, OOP

CERTIFICATIONS & ACHIEVEMENTS

- AWS AI Practitioner Learning Plan (Amazon Web Services, June 2026) — 8 courses: Generative AI, Foundation Models, RAG, Prompt Engineering, Responsible AI, MLOps, AI Security & Governance
- McKinsey Forward Program (McKinsey & Company, Dec 2025) — Structured problem-solving, critical thinking, and adaptability
- Google for Startups — Startup School: Prompt to Prototype (Dec 2025) — AI-powered product development
- BCG GenAI Job Simulation (BCG X, Jul 2025) — Chatbot development and GenAI solution architecture
- AWS APAC Solutions Architecture Simulation (Forage, Jun 2025) — Scalable cloud design using AWS services
- Gold Medalist — International English Olympiad | Ranked Top 1% globally in analytical reasoning